SESSION 2 LECTURE NOTES

Data Fusion: The Pipeline from Raw Sensors to Actionable Intelligence

GeoAI Fusion Bootcamp 2026

Introduction

Geospatial artificial intelligence rests on a pipeline that converts raw observations into decisions with real operational consequences. The pipeline unfolds across five stages: raw data, information, features and indicators, predictions, and actionable decisions. At the first stage, sensors deliver unprocessed observations in large volume, without context or structure. Cleaning and georeferencing this material produces organised, structured datasets. Analysts then derive information, extract patterns, and prepare artificial-intelligence-ready inputs. These inputs feed predictive models that generate risk maps, susceptibility estimates, and probability scenarios. Planners finally translate these outputs into resource allocation, policy design, and measurable real-world impact.

No single sensor captures the complete picture of the Earth's surface. Optical imagery cannot see through persistent cloud cover, a serious limitation in tropical and monsoon-affected regions. Radar struggles to resolve fine surface detail despite penetrating cloud and darkness. Ground-based sensors report accurately, but only from fixed, discrete points on the map. This session addresses the operational problem that follows from these gaps. A missed signal, whether a flood, an infrastructure outage, or a land-use change, carries severe economic and safety consequences.

Data fusion is the discipline that resolves this problem. Fusion reduces blind spots, cancels random noise through redundancy, and produces a statistically more reliable and complete picture than any single input allows. Participants in this session learn the principles of harmonising optical, radar, elevation and climate datasets into a single, structured product suitable for machine learning. This foundation prepares participants for the practical implementation of fusion workflows in Google Earth Engine, covered in Session 4.

Discussion

The Geospatial Pipeline: From Raw Data to Intelligence

The pipeline that structures this session begins with raw data, unprocessed observations collected in large volume without context. The second stage, information, cleans and georeferences these observations into structured datasets. The third stage extracts features and indicators, deriving patterns and artificial-intelligence-ready inputs from the structured information. The fourth stage generates predictions, calculating risk maps, susceptibility estimates, and probability scenarios from the extracted features. The fifth and final stage converts predictions into actionable decisions, covering resource optimisation, policy planning, and measurable real-world impact. Each stage depends entirely on the quality of the stage before it, so an error introduced at the raw-data stage propagates through the entire chain.

The Imperative for Fusion

The problem motivating this session is straightforward. Optical sensors, such as Sentinel-2, miss cloud-covered areas. Synthetic-aperture radar, such as Sentinel-1, struggles with fine detail despite operating regardless of weather. Digital elevation models, such as the Copernicus DEM, capture terrain but not surface change. Climate products, such as ERA5 and CHIRPS, offer high temporal frequency but coarse spatial resolution. Each source covers part of the picture and misses another part. The operational cost of relying on a single source is severe. A missed flood signal, an undetected outage, or an unnoticed land-use change all carry heavy economic consequences. Fusion resolves this by combining sources so that the blind spot in one dataset is filled by the strength of another.

The Multi-Sensor Arsenal

Four broad categories of data anchor most geospatial fusion workflows. Optical sensors, such as Sentinel-2 and Landsat, deliver high visual detail and support land-cover and vegetation monitoring through indices such as the Normalised Difference Vegetation Index. Their blind spot is straightforward, since clouds and darkness block the sensor entirely. Synthetic-aperture radar sensors, such as Sentinel-1 and ALOS, operate in all weather and support surface structure, moisture, and deformation mapping, though their imagery is difficult to interpret due to speckle noise. Digital elevation models, such as TanDEM-X and the Copernicus DEM, provide a highly consistent three-dimensional structural snapshot, supporting slope, aspect, and hydrology modelling, though they cannot capture rapid surface change. Climate reanalysis products, such as ERA5 and CHIRPS, offer excellent temporal frequency and support temperature and drought or flood indicators, though their spatial resolution is coarse, typically between five and thirty-one kilometres.

Bridging the Gap: Space to Ground

A useful analogy for this concept is a team of witnesses describing an accident from different vantage points. Satellite imagery behaves like a witness watching from a fixed, high vantage point on a predictable schedule. It establishes broad spatial context and historical baselines, though its resolution and revisit frequency are fixed by orbital mechanics. Sensor networks, such as ground-based weather stations and air-quality monitors, behave like a witness standing close to the event who reports continuously. They catch rapid, short-lived events that a satellite passing overhead only once every few days would miss. Ground truth, gathered through field surveys and verified observations, behaves like a witness whose account is checked against independent records. It anchors the whole system, since without this calibration reference, systematic model errors go unnoticed and degrade every downstream product.

The Foundation of Fusion: Data Harmonisation

Datasets cannot be fused until they share a single spatial and temporal reality. Harmonisation proceeds through four steps. Georeferencing brings each dataset onto a common coordinate system. Resampling aligns each dataset to the same spatial scale, for instance matching a coarser sensor's pixel grid to a finer one. Temporal alignment synchronises each dataset to the same time frame. Standardisation and quality control unify data formats, handle missing values, and mask clouds. Only once these four steps are complete does an analyst obtain a unified, multi-layer data cube suitable for further processing.

Spatial Harmonisation: Coordinate Systems and Resampling

A dataset referenced to the World Geodetic System 1984 ellipsoid and a dataset projected onto a local Universal Transverse Mercator zone will not align without correction. Subtle georeferencing errors of this kind can shift an entire fused layer by hundreds of metres, invalidating every subsequent machine-learning analysis performed on it. Resampling then addresses the fact that different sensors capture data at different resolutions, such as ten metres for Sentinel-2 against thirty metres for Landsat. Nearest-neighbour resampling is the fastest method and preserves original values, making it appropriate for categorical data, such as land-cover classes. Bilinear and cubic interpolation produce a smoother output by altering the original values, making them appropriate for continuous data, such as elevation or temperature. As a general rule, aggregating finer data to a coarser resolution preserves statistical validity more reliably than the reverse.

Temporal Harmonisation: Synchronising the Clocks

Data sources are rarely captured at the exact same moment. For fast-changing conditions, such as wildfires or floods, even a mismatch of a few hours can meaningfully distort a fused output. Three strategies address this gap. The nearest-available strategy selects the closest temporal acquisition to the target date. Interpolation mathematically bridges the gap between two time points. Temporal aggregation creates composites at daily, monthly, or seasonal intervals, condensing large observation volumes, reducing cloud-related noise, and revealing long-term seasonal dynamics.

Data Quality Guardrails

Fusing poor-quality data amplifies errors rather than reducing them, an outcome that undermines the entire purpose of the exercise. Three guardrails protect against this outcome. Accuracy checks compare dataset values against independent ground-truth references, such as field surveys, catching invisible biases before they propagate further. Completeness management addresses missing data explicitly, for instance cloud-obscured pixels, through exclusion, neighbour interpolation, or substitution from a secondary source. Consistency resolution establishes rules for when sources disagree, such as trusting the higher-resolution source or flagging the discrepancy for manual review.

The Dividends of Fusion

Combining sources produces a picture that is more accurate, more complete, and more reliable than any single source could deliver alone. Fusion eradicates blind spots. Where optical imagery misses cloud-covered zones and radar struggles with fine detail, the combined product pieces together the fragments each source captured well. Fusion also cancels noise, since redundancy across sensor modalities filters out random anomalies and sensor-specific artefacts. Finally, fusion creates statistical reliability, since a combined estimate narrows the confidence interval around a prediction, a critical benefit when a missed signal carries a heavy operational cost.

Constructing the 4D Feature Cube

The pre-fusion synthesis stage transforms disparate observations into a structured data cube spanning space, time, and spectral bands. Perfect spatial and temporal alignment integrates multi-source intelligence efficiently into this structure. The resulting well-organised mathematical object is the non-negotiable foundation for powerful machine-learning models and reliable predictive geospatial insight.

Three Levels of Fusion

Fusion can occur at three distinct levels, and the appropriate level depends on how far each source has already been processed. Pixel-level fusion merges raw sensor values directly, for instance stacking spectral bands from two different satellite passes before any interpretation occurs. Feature-level fusion combines derived attributes, such as edges, textures, and objects, after each source has been independently processed and categorised. Decision-level fusion merges the final outputs of separate classifiers or models, and it is the most robust choice when the underlying source data formats differ substantially.

Conclusion

This session established data fusion as the discipline that converts fragmented, single-sensor observations into a complete, statistically reliable, and machine-learning-ready product. Harmonisation, spatial and temporal alignment, and rigorous quality control together determine whether a fused dataset strengthens or undermines downstream analysis. The four-dimensional feature cube, built from harmonised optical, radar, elevation, and climate layers, represents the practical endpoint of this process.

Participants should now recognise why no single sensor suffices for operational decision-making, and which of the three fusion levels, pixel, feature, or decision, suits a given combination of sources. Session 4 builds directly on this foundation, demonstrating how these principles are implemented in practice using Google Earth Engine.